✅ 1. Competency #2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

- Highest relevance.
- The entire task hinges on evaluating the reasonableness of the tester's claim in light of the data.
- Students must:
  - Compute actual faulty units:
    - Video: 2,000 × 5% = 100
    - Audio: 6,000 × 3% = 180
  - Compare: 100 < 180 → claim is false
- Crucially, it's not enough to calculate correctly; they must judge whether the conclusion ("more video faulty") aligns with reality.
- This directly embodies competency #2: assessing if a stated conclusion makes sense in context.

✅ 2. Competency #7: Construct and communicate explanations and arguments in the context of the problem

- Highly relevant (explicitly required).
- The prompt demands: "Give a mathematical argument to support your answer."
- A strong response must:
  - State the claim is incorrect
  - Show calculations clearly
  - Explain in context: "Although the fault *rate* is higher for video players (5% > 3%), the much larger production volume of audio players (6,000 vs. 2,000) results in more audio players being faulty overall."
- This is a contextualized mathematical justification, not just a numerical answer—precisely what #7 emphasizes.

✅ 3. Competency #1: Interpret a mathematical result back into the real-world context

- Highly relevant.
- After computing 100 vs. 180, students must reconnect these numbers to the real-world scenario:

> "180 audio players are sent for repair each day, which is more than the 100 video players."

- This step transforms abstract numbers into meaningful statements about company operations, fulfilling competency #1.

✅ 4. Competency #6: Use knowledge of how the real world impacts outcomes to make contextual judgments about how results should be applied

- Moderately to highly relevant.
- The problem tests understanding that absolute numbers matter more than percentages in this context (repair workload depends on *how many* units, not *what proportion*).
- Students must recognize that the tester confused rate with quantity—a common real-world misinterpretation.
- Using this insight to reject the claim demonstrates applying real-world logic to adjust or critique a naive mathematical interpretation, which is the essence of #6.

🟡 5. Competency #3: Identify the limits of the model used to solve a problem

- Weak but present relevance.
- The model assumes:
  - Percentages represent exact daily averages
  - No variation in production or fault rates
  - All faults are detected and reported
- While the task doesn't require discussing these limitations, a sophisticated student might note:

> "This conclusion holds on average, but daily numbers may vary."

- However, since the problem explicitly uses "on average," this competency is only marginally engaged.

✅ 1. 能力 #2：判断数学结果或结论在问题情境中是否合理

> （Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem）

- 最高相关性。
- 整道题的核心就是评估测试员说法的合理性。
- 学生需：
  - 计算实际故障数量：
    - 视频播放器：2,000 × 5% = 100
    - 音频播放器：6,000 × 3% = 180
  - 比较：100 < 180 → 说法错误
- 关键在于，不能只算对数字，还必须判断“视频故障更多”这一结论是否符合实际情况。
- 这正是能力 #2 的核心：在具体情境中判断一个结论是否合理。

✅ 2. 能力 #7：在问题情境中构建并表达解释与论证

> （Construct and communicate explanations and arguments in the context of the problem）

- 高度相关（题目明确要求）。
- 题干要求：“给出一个数学论证来支持你的答案。”
- 优秀回答应包含：
  - 明确指出说法不正确
  - 清晰展示计算过程
  - 在情境中解释：“虽然视频播放器的故障率更高（5% > 3%），但由于音频播放器的日产量远高于视频播放器（6,000 vs. 2,000），因此每天故障的音频播放器总数反而更多。”
- 这是一种结合情境的数学说理，而非仅给出数值答案——这正是能力 #7 所强调的。

✅ 3. 能力 #1：将数学结果还原到现实世界情境中

> (Interpret a mathematical result back into the real-world context)

- 高度相关。
- 在得出 100 与 180 后，学生必须将这些数字重新联系到现实场景：

> “每天有 180 台音频播放器送去维修，比视频播放器的 100 台更多。”

- 这一过程将抽象数字转化为对公司运营有意义的陈述，完全契合能力 #1。

✅ 4. 能力 #6：利用现实世界知识对结果进行情境化判断，以决定如何应用结果

> (Use knowledge of how the real world impacts outcomes to make contextual judgments about how results should be applied)

- 中高相关性。
- 本题考查学生是否理解：在此情境下，绝对数量比百分比更重要（维修工作量取决于“多少台”，而非“占比多少”）。
- 学生需意识到，测试员混淆了“比率”与“数量”——这是现实中常见的误判。
- 能够运用这一洞察来反驳错误说法，正体现了能力 #6：用现实逻辑调整或批判表面化的数学解读。

🟡 5. 能力 #3：识别用于解决问题的模型的局限性

> (Identify the limits of the model used to solve a problem)

- 弱相关但存在。
- 本题所用模型隐含假设：
  - 百分比代表精确的日均值
  - 生产量和故障率无波动
  - 所有故障均被检测并送修
- 虽然题目未要求讨论这些限制，但高水平学生可能会指出：

> “该结论在‘平均’意义上成立，但每日实际数量可能有波动。”

- 然而，由于题目明确使用“平均而言”，此项能力仅轻微涉及。